

Solution : Let m_1 and m_2 be the masses of the spheres and u be their velocities (in opposite directions)

One of the spheres comes to rest after impact, and let the velocity of the other sphere be v after impact.

Then by Newton's law on impact

$$0 - v = e(u + u) = 2ue$$

$$\text{i.e., } v = -2eu$$

Also, by the principle of conservation of linear momentum

$$m_1 v = m_1 u - m_2 u$$

$$\text{or, } (1 + 2e)m_1 u = m_2 u$$

$$\therefore m_1 : m_2 = 1 : (1 + 2e)$$

Ex. 6. A heavy elastic ball drops from the ceiling of a room, and after rebounding twice from the floor reaches a height equal to one-half that of the ceiling. Find the coefficient of restitution.

Solution : Let h be the height of the ceiling of the room and e , the coefficient of restitution.

The height to which the ball rebounds after the first impact with the floor $= e^2 h$

and the height to which the ball rises after the second rebound $= e^2 \cdot e^2 h = e^4 h$.

By the condition of the problem,

$$e^4 h = \frac{1}{2} h \quad \text{or, } e^4 = \frac{1}{2}$$

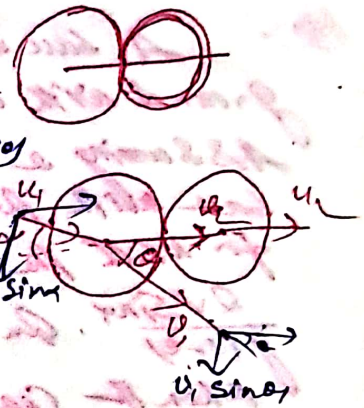
$$\therefore e = \sqrt[4]{\frac{1}{2}}$$

Ex. 7. A ball of mass 5 oz. is dropped from a height of 16 ft and after hitting the ground it rebounds to a height of 9 ft. Find the impulsive force exerted by the ground, if contact with the ball lasts for $\frac{1}{100}$ second.

Q:- A smooth billiard ball impinges on another ball (at rest) of same mass and radius in a direction that makes an angle α with the line of centres at the moment of impact. And e is their coefficient of restitution. Prove that the angle through which the direction of motion of the impinging ball is deviated is

$$\tan^{-1} \left[\frac{(1+e) \tan \alpha}{1-e+2 \tan^2 \alpha} \right]$$

* Let m be the mass of each ball.
Let u be the velocity of first billiard ball just before impact. And let v_1 and v_2 be the velocities of the two balls just after impact in the directions making an angle θ_1 and θ_2 along the line of centre resp.



Since the components of velocity perpendicular to the line of impact of each the first billiard ball remains unchanged.

$$u \sin \alpha = v_1 \sin \theta_1 \quad \text{--- (1)}$$

By Newton's experimental law

$$u \cos \alpha = v_1 \cos \theta_1$$

$$v_2 - v_1 \cos \theta_1 = e(u \cos \alpha - 0) = eu \cos \alpha \quad \text{--- (2)}$$

By the principle of conservation of linear momentum

$$mv_2 + mv_1 \cos \theta_1 = mu \cos \alpha$$

$$\text{or } v_2 + v_1 \cos \theta_1 = u \cos \alpha \quad \text{--- (3)}$$

Subtract (2) from (3)

$$2v_1 \cos \theta_1 = u \cos \alpha (1-e)$$

$$\text{or } v_1 \cos \theta_1 = \frac{1}{2} (1-e) u \cos \alpha \quad \text{--- (4)}$$

Dividing (4) by (1)

$$\tan \theta_1 = \frac{1}{2} (1-e) \tan \alpha$$

If ϕ be the angle through which the direction of motion is deviated, then

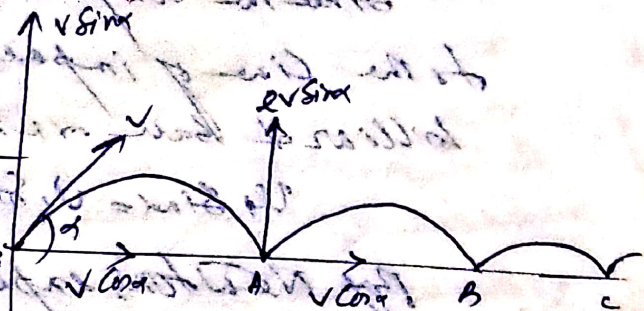
$$\phi = \theta_1 - \alpha$$

$$\tan \phi = \tan(\theta_1 - \alpha) = \frac{\tan \theta_1 - \tan \alpha}{1 + \tan \theta_1 \tan \alpha} = \frac{\frac{2 \tan \alpha}{1-e} - \tan \alpha}{1 + \frac{2 \tan \alpha \tan \alpha}{1-e}} = \frac{(1+e) \tan \alpha}{1-e+2 \tan^2 \alpha}$$

$$\phi = \tan^{-1} \left[\frac{(1+e) \tan \alpha}{1-e+2 \tan^2 \alpha} \right]$$

Ex:- A ball is thrown from a point on a smooth horizontal ground with a velocity v at angle α to the horizon. Assuming e to be the coefficient of restitution of the ball with the ground, show that the total time for which the ball remains on the ground is $\frac{2v \sin \alpha}{g(1-e)}$, and that its distance from the starting point when it ceases to rebound is $\frac{v^2 \sin 2\alpha}{g(1-e)}$.

Ans. Let the ball be projected from a point O with velocity v at angle α to the horizon. Then the ball describes a parabolic orbit and then it strikes the ground with the same velocity v .



at angle α with the horizontal and rebounds.

The horizontal and vertical components of v are $v \cos \alpha$ and $v \sin \alpha$ resp. The ball will describe a series of parabolic orbits until it ceases to rebound.

The horizontal component $v \cos \alpha$ remains unchanged after impact, but the vertical component will change. The vertical component after first, second, third, ... etc. rebounds become $v \sin \alpha$, $e v \sin \alpha$, $e^2 v \sin \alpha$, ... etc.

If $T_1, T_2, T_3, \dots$ etc be the times of describing the first, second, third, ... etc parabolic orbits, resp. Then

$$T_1 = \frac{2V \sin \alpha}{g}, T_2 = \frac{2V \sin \alpha}{g}, T_3 = \frac{2V \sin \alpha}{g}, \dots \text{etc.}$$

Hence the required total time for which the ball is in the air

$$= T_1 + T_2 + \dots$$

$$= \frac{2V \sin \alpha}{g} + \frac{2V \sin \alpha}{g} + \frac{2V \sin \alpha}{g} + \dots$$

$$= \frac{2V \sin \alpha}{g} (1 + 1 + 1 + \dots)$$

$$= \frac{2V \sin \alpha}{g} \cdot \frac{1}{1-r} = \frac{2V \sin \alpha}{g(1-r)}$$

The required distance

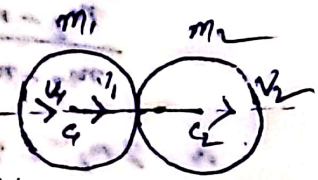
= the horizontal distance travelled during

$$\text{the total time } \frac{2V \sin \alpha}{g(1-r)}$$

$$= V \cos \alpha \cdot \frac{2V \sin \alpha}{g(1-r)} = \frac{2V^2 \sin \alpha \cos \alpha}{g(1-r)}$$

Ex: A ball A, of mass m_1 lb impinges directly on another ball B, of mass m_2 lb which is at rest. After impact, B impinges directly on a third ball C of mass m_3 lb which is also at rest. If the velocity imparted to C is the same as A had at first and if the balls are perfectly elastic. Show that $(m_1 + m_2)(m_2 + m_3) = 4m_1 m_2$.

Ans. Let the ball A of mass m_1 lb impinge directly on another ball of mass m_2 lb which is at rest with a velocity u . Let v_1 & v_2 be the respective velocities after impact along the line of centres. Then by Newton's experimental law



$$v_2 - v_1 = e(u - 0) \quad \text{where } e = 1$$

$$\text{or } v_2 = u + v_1 \quad \text{--- (1)}$$

And by the principle of Conservation of Linear momentum.

$$m_2 v_2 + m_1 v_1 = m_1 u_1 \quad (2)$$

Now let the v_3 & v_4 be the respective velocities of the balls b & c after impact along the line of contact.

Now let the ball b with velocity v_2 impinge the ball c , at rest directly and let v_3 & v_4 be their respective velocities after impact along the line of impact.

By Newton's experimental law,

$$v_4 - v_3 = e v_2 \quad (3)$$

$$\text{since } v_4 - v_3 = v_2 \quad (3) \text{ and } v_4 = u_4$$

And by the principle of Linear momentum

$$m_3 v_4 + m_2 v_3 = m_2 v_2$$

$$\text{given } m_3 u_4 + m_2 v_3 = m_2 v_2 \quad (4)$$

Now from (3) & (4)

$$u_4 - v_3 = u_4 + v_1$$

$$\text{or, } v_3 = -v_1 \quad (5)$$

$$(4) \Rightarrow m_3 u_4 + m_2 v_3 = m_2 (u_4 + v_1) \quad \text{by } (5)$$

$$\text{or, } (m_3 - m_2) u_4 = m_2 (v_1 - v_3) = 2 m_2 v_1 \quad (6)$$

$$(2) \Rightarrow m_2 (u_4 + v_1) + m_1 v_1 = m_1 u_4$$

$$\text{or, } u_4 (m_2 - m_1) = - (m_1 + m_2) v_1 \quad (7)$$

Dividing (6) by (7)

$$\frac{m_3 - m_2}{m_3 - m_2} = \frac{-(m_1 + m_2)}{2 m_2}$$

$$\text{or, } 2 m_2 v_1 - 2 m_1 v_1 = - m_1 m_3 - m_2 m_3 + m_1 m_2 + m_2^2$$

$$\text{or, } m_1^2 + m_2 m_3 + m_1 m_3 + m_1 m_2 = 4 m_1 m_2$$

$$\text{or, } m_2 (m_1 + m_3) + m_1 (m_2 + m_3) = 4 m_1 m_2$$

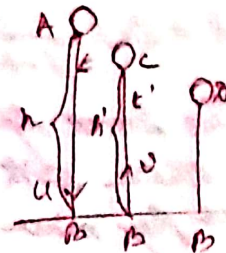
$$\text{or, } (m_1 + m_3) (m_1 + m_2) = 4 m_1 m_2$$

Q. 253 A ball falls from a height upon a fixed horizontal plane. Show that the whole distance travelled before the ball finished rebounding is $\frac{1+e}{1-e} \cdot h$ and the time taken is $\sqrt{\frac{2h}{g}} \left(\frac{1+e}{1-e} \right)$.

Ans. Let u be the velocity acquired by the ball on reaching the fixed horizontal floor at B when it is dropped from a height h above the floor. Then

$$u^2 = 0 + 2gh \quad [v^2 = u^2 + 2fs]$$

$$\text{or } u = \sqrt{2gh}$$



Here u is the velocity of approach just before impact.

Let v be the upward velocity of the ball at B immediately after impact with the floor. Then v is the velocity of separation just after impact. But u and v are along the common normal at the point of contact of the ball and floor.

By Newton's experimental law of impact, we have

$$v = eu$$

$$\text{or } v = e\sqrt{2gh}$$

If h' be the height to which the ball rebounds and $BC = h'$, then the ball comes to rest at C at a height h' above the floor and we have

$$0 = v^2 - 2gh' \quad \text{or } h' = \frac{v^2}{2g} = \frac{e^2 \cdot 2gh}{2g} = e^2 h$$

Thus, for the first rebound, height descended $= h' = e^2 h$

for the second rebound, height descended $= e^2 h' = e^2 \cdot e^2 h = e^4 h$

And so on, $\therefore$ height descended $= e^4 h' = e^4 \cdot e^2 h = e^6 h$

And so on.

Total distance travelled

$$= AB + 2BC + 2BD + \dots$$

$$= h + 2e^2 h + 2e^4 h + \dots$$

$$= h + 2e^2 h (1 + e^2 + e^4 + \dots)$$

$$= h + 2e^2 h \cdot \frac{1}{1-e^2} = \frac{h - e^2 h + 2e^2 h}{1-e^2} = \frac{h(1+e^2)}{1-e^2}$$

If t be the time from A to B, then $u = gt$ and $h = \frac{1}{2}gt^2$ or $t = \sqrt{\frac{2h}{g}}$

If t' be the time from B to C, then

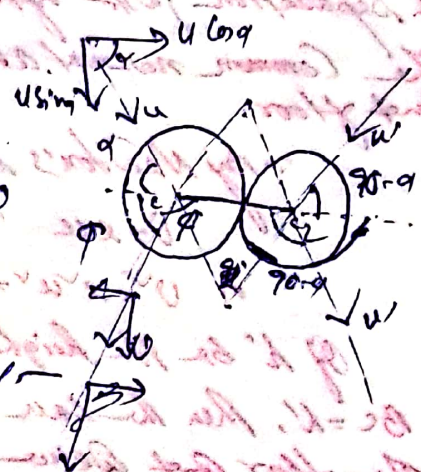
$$0 = v - gt' \quad \text{or } t' = \frac{v}{g} = \frac{e\sqrt{2gh}}{g} = \frac{e\sqrt{2h}}{\sqrt{g}} = et = e\sqrt{\frac{2h}{g}}$$

Then, for the first rebound, $t' = u \cdot t' = 2t$ and soon
 Hence the total time of describing the whole distance
 $= t + 2t' + 2t' + \dots = t + 2t + 2t + 2t + \dots$
 $= t + 2t (1 + 1 + 1 + \dots)$
 $= t + 2t \cdot \frac{1}{1-e} = \frac{t - et + 2et}{1-e} = \frac{(1+e)t}{1-e} = \frac{1+e}{1-e} \sqrt{\frac{2h}{g}}$

Let two smooth spheres of masses m and m' move towards each other with velocities u and u' at right angles. After impact they move away at right angles. If α is the angle that u makes with the line of action from that the coefficient of restitution e is given by

$$e = \frac{m u \cos \alpha + m' u' \sin \alpha}{m m' (u \cos \alpha + u' \sin \alpha)}$$

Ans. Let V and $0'$ be the velocity of the 1st & 2nd ball just after impact & let the 1st sphere move at an angle ϕ with the line of centers.



Since the components of velocity perpendicular to the line of impact of each ball remains unchanged.

$$1. u \sin \alpha = V \sin \phi \quad \text{--- (1)}$$

$$u' \cos \alpha = V' \cos \theta \quad \text{--- (2)}$$

By Newton's experimental law

$$V' \sin \theta + V \cos \phi = e (u \cos \alpha + u' \sin \alpha) \quad \text{--- (3)}$$

By the principle of conservation of linear momentum

$$m u \cos \alpha - m' u' \sin \alpha = -m V \cos \phi + m' V' \sin \theta$$

$$= -m \frac{u \sin \alpha \cos \phi}{\sin \phi} + m' \frac{u' \cos \theta \sin \theta}{\cos \theta} \quad \text{by (1) \& (2)}$$

$$\text{or, taking } m' u' \cos \theta - \text{fancy } (m u \cos \alpha - m' u' \sin \alpha) - m u \sin \alpha = 0$$

$$\therefore \text{fancy} = \frac{m u \cos \alpha - m' u' \sin \alpha + m u \sin \alpha}{m' u' \cos \theta}$$

$$= \frac{m u \cos \alpha - m' u' \sin \alpha + m u \sin \alpha + m' u' \sin \theta}{2 m' u' \cos \theta} = \frac{m u}{m' u'}$$

